



Fig. 5: India's Param Yuva II supercomputer
(Source: www.crazyengineers.com)

on software called Local Analysis and Prediction System (LAPS) that can process up to a million separate pieces of information each day. This entire system consists of several hardware and software components in an integrated environment, namely, a high-performance computer system (IBM RS/6000 SP), a forecasting model (such as RAMS, MM5 or WRF), a data-assimilation package (like LAPS), visualisation software (Data Explorer) and associated peripherals.

LAPS software developed by Forecast Systems Laboratory (FSL) of NOAA is a data assimilation and analysis package, which takes in local, national as well as global data from various sources such as satellite, radar and aircraft. LAPS functions as a pre-processing assimilation step, therefore resultant grid data from LAPS is used as initial conditions for the model. It executes serially on a single processor and does more than just model initialisation.

LAPS provides a high-resolution view of the atmosphere in its current state and derived products (like icing, visibility and clouds) and variables (like heat index and buoyancy), which prove useful for a wide array of real-time applications. It produces surface analysis and three-dimensional (3D) wind, temperature, cloud and moisture analyses, and also incorporates facilities to assess data quality.

Deep Thunder's power can be

gauged by the fact that it can produce highly-accurate weather predictions within the narrow ranges of a single city. The system was used during 1996 Atlanta Olympics to successfully schedule weather-affected events like sailing and the closing ceremony. It is also proposed to be used for 2016 Summer Olympics in Rio de Janeiro, Brazil.

How India forecasts its weather

Param was the first mission taken up by C-DAC for the development of a high-performance parallel computer and was completed in July 1991. Later, Param Yuva II was launched in 2013, which was capable of performing at a peak of 524 teraflop per second and was used for research in weather forecasting and seismic data analysis.

In 2015, India achieved another milestone with the release of supercomputer Bhaskara. This supercomputer assists meteorologists in research and predicts the weather, which includes effective forecast of tropical cyclones, heavy rainfall and cloud-burst events. Bhaskara enables Earth System Science Organisation-National Centre for Medium Range Weather Forecasting (ESSO-NCMRWF) to make very high-resolution 10-day deterministic weather forecasts and probabilistic forecasts from a 44-member ensemble prediction system. This is within the generally accepted time window of about five hours from the standard observation time with a horizontal resolution of 1.5km and probabilistic forecasts using an ensemble prediction system.

Bhaskara is powered with IBM iDataPlex supercomputer that has a peak computing power of 350 teraflops with 67 terabytes of aggregate memory. This takes the total ESSO high-performance computing facility to a peak computing power of 1.14 petaflops.

Conclusion

The tsunami disaster that devastated the world a few years ago could have been less lethal had better tsunami-warning systems been in place. Scientists studying tsunamis have their computers working as fast as possible, but they still cannot quite figure out how the killer waves behave or predict when and where these will strike next.

Computer modelling of tsunamis is very complex as it involves solving the full 3D motion of water. However, by using computer models of wave motions based on the known terrain of the undersea floor, scientists have managed to generate tentative maps that display the havoc a big tsunami could cause.

The IT industry needs to take up tsunamis forecasting based on effective computer modelling as an opportunity. In 2010, the storm and mudslides in Rio de Janeiro killed more than 200 people and rendered thousands homeless. This calamity made the city embrace IBM Deep Thunder weather-forecasting system in order to be better geared up for such future disasters.

Apart from saving lives, weather forecasting has a number of applications and uses. For example, public utility companies that provide, say, electricity can benefit vastly from forecasts. This can help identify areas where incoming storms are likely to damage poles, transformers and power lines. Hence, the required number of maintenance staff can be quickly transported to areas near disaster sites to carry out repair work and radically decreasing downtime.

In the area of agriculture, sophisticated weather-forecasting supercomputing systems can decide the best times to plant, irrigate and harvest crops, based on dynamic weather conditions of the various farm locations. This is likely to result in better crop prices, less water and manpower wastage for farmers. **EFY**